

Jiamin Yao

Greater St. Louis Area (open to relocation) | 618-631-5638 | jamiyao0508@gmail.com | [LinkedIn](#) | [Portfolio](#)

SKILLS

Languages: Python, SQL, JavaScript, TypeScript, C#, C/C++, R, Java, Bash

AI/ML: PyTorch, TensorFlow, scikit-learn, Transformers, OpenAI API, Model Calibration/Quantization/Fine-tuning

Data Engineering: Oracle SQL, SAP Workflows, Docker, Cloudflare, Google Cloud Run, Firebase, Supabase

Frameworks & Tools: React, SvelteKit, Next.js, Node.js/Express, Git, Agile/Scrum, Power BI, MATLAB

SELECTED PUBLICATIONS

SPQ: An Ensemble Technique for Large Language Model Compression | *First Author* | LREC 2026 Oral Presentation

- Engineered a novel ensemble compression methodology that achieved a **75%** reduction in memory footprint and a **1.9x** inference speedup. Successfully maintained robust LLM performance across zero-shot downstream benchmarks including C4, TruthfulQA, and GSM8K.

A New Technique for LLM Quantization | Under Review for NeurIPS 2026

A New Technique for Diffusion Model Compression | Targeting AAAI 2027

WORK EXPERIENCE

AI Research Assistant | *SIUE* (Edwardsville, IL) | Feb 2026 – Present

- Researched and implemented quantization methods to compress LLMs and diffusion models.
- Built model retraining and fine-tuning pipelines to optimize quantized models and test them on downstream benchmarks.

Software Developer | *AierTeam* (St. Louis, MO) | May 2025 – Aug 2025

- Built an AI agent using C#, Semantic Kernel, and LangGraph to automatically generate and execute multi-step tasks.
- Developed a GUI automation tool with operations like keystroke injection, and used Docker for environment isolation.
- Worked in an Agile team environment, using Git for version control, code reviews, and CI/CD workflows.

Advisor Global Buyer | *Dell Technologies* (Xiamen, China) | Jul 2019 – Sep 2023

- Built Oracle/SQL ETL pipelines to automate supply chain risk assessments, which reduced response time by 75%.
- Optimized SAP workflows using SQL queries, improving data allocation accuracy by 15% and removing duplicate data.
- Created Power BI models and automated dashboards, reducing manual reporting time by 82%.

TECHNICAL PROJECTS

CS Deadline Tracker (Full-Stack / DevOps)

- Built a high-performance, timezone-aware static web application tracking 800+ global CS conference deadlines.
- Implemented automated web-scraping scripts (WikiCFP/CSRankings) for real-time data aggregation, integrated live countdown tickers, and engineered one-click calendar exports (iCal/Google Calendar); deployed via Cloudflare Pages.

Draw a Mushroom (Full-Stack / Real-Time Game)

- Developed a multiplayer drawing game using vanilla JavaScript, Node.js/Express, and Firebase Firestore.
- Built a real-time physics canvas, a global community voting engine, and a live leaderboard; containerized and deployed across Cloudflare Workers and Google Cloud Run.

Gongshami (AI Chatroom / Web Application)

- Designed an interactive web using SvelteKit and Pocketbase for secure user authentication and real-time state storage.
- Integrated customized AI agent personalities, enabling dynamic context-aware user chats; deployed via Docker.

Chatbot Preference Prediction Pipeline (Machine Learning)

- Constructed an end-to-end evaluation pipeline comparing LightGBM, CNN, BERT, and Gemma architectures to predict and rank user chatbot preference alignment.

EDUCATION

Master of Science in Computer Science | Southern Illinois University Edwardsville | GPA: **3.7** | Jan 2024 – Dec 2025

Coursework: Machine Learning, Deep Learning, Artificial Intelligence, Digital Image Processing, Advanced Algorithms

Bachelor of Science in Marine Technology, Xiamen University | Sep 2015 – Jun 2019

Coursework: Fundamental Computer Application, C Programming, Data Analysis